Enable TDE

Oracle MAA best practice recommends using Oracle Transparent Data Encryption (TDE) to encrypt both primary and standby databases to ensure all data is encrypted at-rest. Data can be converted during the migration process but it's highly recommended to convert to TDE prior to migration to provide the most secure Data Guard environment.

1. Switch to the **oracle** user.

sudo su – oracle

2. Connect to the database.

sqlplus / **as** sysdba

From the primary:

```
[opc@primary ~]$ sudo su - oracle
Last login: Thu Apr  9 15:36:38 GMT 2026 on pts/0
[oracle@primary ~]$ sqlplus / as sysdba

SQL*Plus: Release 19.0.0.0.0 - Production on Thu Apr 9 16:10:54 2026
Version 19.20.0.0.0

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Connected to:
Oracle Database 19c Enterprise Edition Release 19.0.0.0.0 - Production
Version 19.20.0.0.0
```

3. Set Wallet_ROOT static init parameter.

alter system set wallet_root='/u01/app/oracle/product/19c/dbhome_1/wallet' **scope**=spfile;

From the primary:

```
SQL> alter system set wallet_root='/u01/app/oracle/product/19c/dbhome_1/wallet' scope=spfile;
System altered.
SQL>
```

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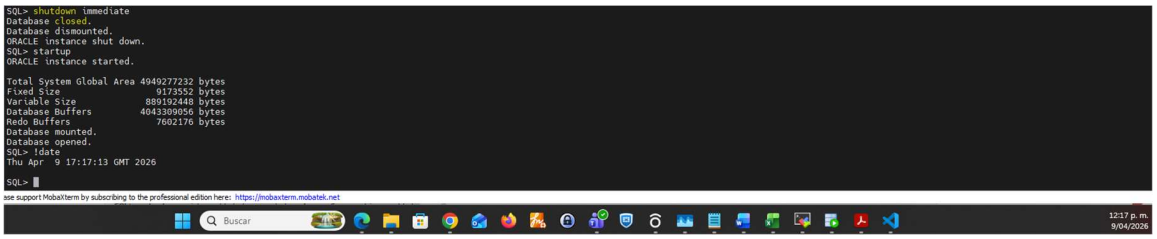
4. Restart the database to enable the parameter.

shutdown immediate

startup

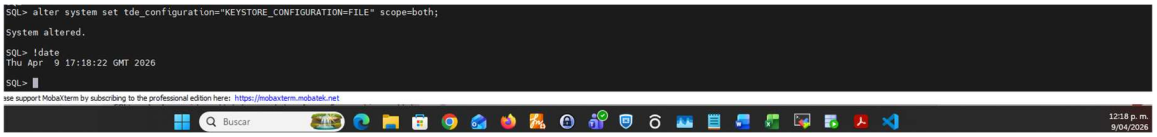
From the primary:



5. Set TDE_CONFIGURATION parameter to set the keystore type to file.

alter system set tde_configuration="KEYSTORE_CONFIGURATION=FILE" **scope=both**;

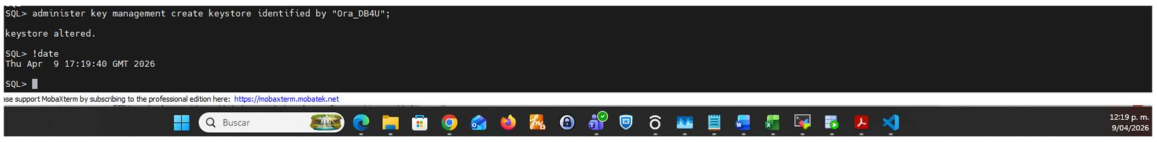
From the primary:



6. Created a password-protected keystore.

SQL> administer key management **create** keystore identified **by** "Ora_DB4U";
keystore altered.

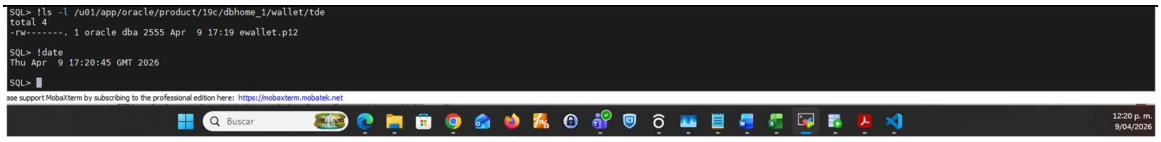
From the primary:



7. Check the keystone files. ewallet.p12 is the password-protected keystore.

SQL> !ls -l /u01/app/oracle/product/19c/dbhome_1/wallet/tde
total 4
-rw-------. 1 oracle dba 2555 Sep 25 05:23 ewallet.p12

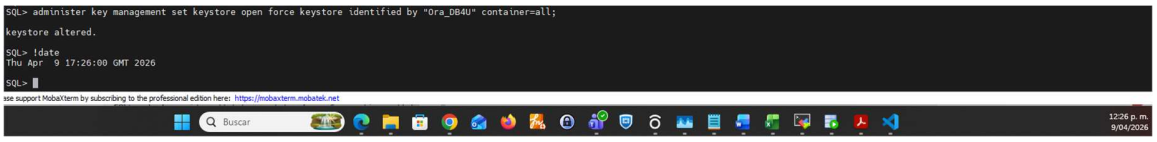
From the primary:



8. Open the password-protected keystore.

SQL> administer key management **set** keystore **open** force keystore identified **by** "Ora_DB4U" container=**all**;
keystore altered.

From the primary:

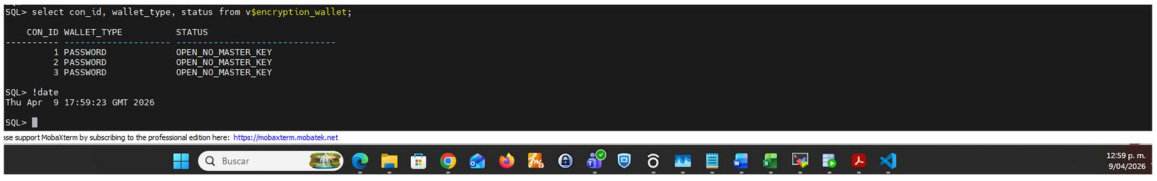


9. Check current status of the wallet.

SQL> **select** con_id, wallet_type, status **from** v$encryption_wallet;

```
 CON_ID WALLET_TYPE       STATUS
---------- -------------------- ------------------------------
     1 PASSWORD       OPEN_NO_MASTER_KEY
     2 PASSWORD       OPEN_NO_MASTER_KEY
     3 PASSWORD       OPEN_NO_MASTER_KEY
```

From the primary:

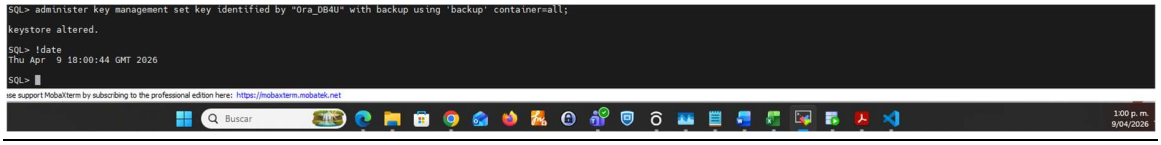


10. Create master key.

    SQL> administer key management **set** key identified **by** "Ora_DB4U" **with** backup **using** 'backup' container=**all**;

    keystore altered.

From the primary:



11. Verify the keystore, you can see the wallet is opened by password.

    SQL> **select** con_id, wallet_type, status **from** v$encryption_wallet;

```
     CON_ID WALLET_TYPE       STATUS
---------- -------------------- ------------------------------
         1 PASSWORD             OPEN
         2 PASSWORD             OPEN
         3 PASSWORD             OPEN
```

From the primary:

```
SQL> select con_id, wallet_type, status from v$encryption_wallet;

    CON_ID WALLET_TYPE          STATUS
---------- -------------------- ------------------------------
         1 PASSWORD             OPEN
         2 PASSWORD             OPEN
         3 PASSWORD             OPEN

SQL> !date
Thu Apr  9 18:04:07 GMT 2026

SQL>
```

12. Make the Keystore Auto-Login.

SQL> administer key management **create** auto_login keystore **from** keystore identified **by** "Ora_DB4U";

keystore altered.

From the primary:

```
SQL> administer key management create auto_login keystore from keystore identified by "Ora_DB4U";

keystore altered.

SQL> !date
Thu Apr  9 18:13:57 GMT 2026

SQL>
```

13. To start using the auto-login keystore, we should close the password-protected keystore.

SQL> administer key management **set** keystore **close** identified **by** "Ora_DB4U" container=**all**;

keystore altered.

From the primary:

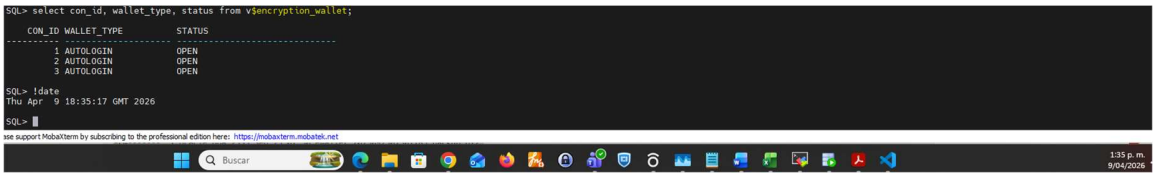
```
SQL> administer key management set keystore close identified by "Ora_DB4U" container=all;

keystore altered.

SQL> !date
Thu Apr  9 18:32:49 GMT 2026

SQL>
```

14. Verify the keystore again, now you can see the wallet is configure to autologin.

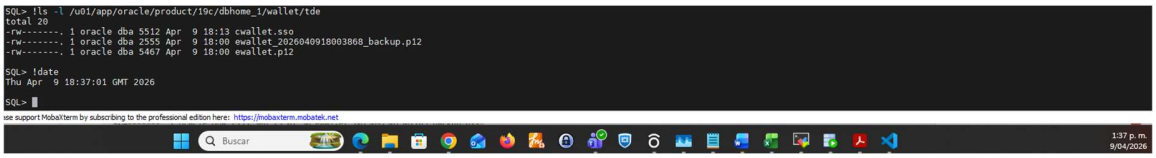
SQL> **select** con_id, wallet_type, status **from** v$encryption_wallet;

CON_ID	WALLET_TYPE	STATUS
1	AUTOLOGIN	**OPEN**
2	AUTOLOGIN	**OPEN**
3	AUTOLOGIN	**OPEN**

From the primary:



15. We can check what files we have now. In which, ewallet.p12 is the password-protected keystore, cwallet.sso is the auto-login keystore and ewallet_*_backup.p12 is the keystore backup file.

SQL> !ls -l /u01/app/oracle/product/19c/dbhome_1/wallet/tde
total 20
-rw-------. 1 oracle dba 5512 Sep 25 07:31 cwallet.sso
-rw-------. 1 oracle dba 2555 Sep 25 07:30 ewallet_2023092507301183_backup.p12
-rw-------. 1 oracle dba 5467 Sep 25 07:30 ewallet.p12

From the primary: